

Sourik Dutta

(407) 791-9955 | sd5913@nyu.edu | NY | [Linkedin](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Systems & Backend Software Engineer with 2+ years of experience engineering high-throughput distributed backend systems, real-time telemetry pipelines, and resilient cloud infrastructure. Proven track record sustaining **99.86% SLO at 3,400+ req/sec**, scaling Kafka streaming pipelines, and automating incident triage. Expert across **Java (Spring Boot), C++, Python (FastAPI), Go, and C# .NET**, with deep proficiency in **PostgreSQL, Redis, Kafka, Docker, and Kubernetes**. MS in Computer Science from NYU (2026).

EDUCATION

Master of Science in Computer Science
New York University

05/2026
New York City

Relevant Coursework: Algorithms, Cloud Computing and Big Data, Machine Learning , Java , Internet Security, Software Engineering, Blockchain

Bachelor of Technology in Computer Science
Amity University

08/2022
Noida

Relevant Coursework: Data Structures, Databases, Web Development, Operating Systems

WORK EXPERIENCE

Software Engineer - Cloud & AI Platform Engineering
Insight Enterprises

Jan. 2023 – Nov. 2023
Remote

- Architected a production LLM-as-a-service platform on Azure, deploying Java Spring Boot microservices on Azure Kubernetes Service (AKS) with Resilience4j circuit breaking, Redis prompt caching, and low-latency OpenAI API integration, scaling reliable AI tooling to 13,000+ internal teammates.
- Built an AIOps telemetry ingestion pipeline on Azure Event Hubs feeding real-time signals into Z-Score and EWMA anomaly models on AKS, engineering observability tracking latency, token usage, and detection accuracy to automate pre-escalation triage and route high-severity incidents across 3 microservices.
- Sustained a 99.86% SLO at 3,412 concurrent req/sec across 12,143 signals by root-causing high-severity HttpClientFactory TIME_WAIT socket leaks under heavy load, repartitioning Kafka clusters from 12 to 48 partitions, and codifying DRI incident runbooks as version-controlled decision trees
- Delivered an Angular 15 real-time operations dashboard with WebSocket feeds via Azure SignalR, OnPush change detection maintaining 60fps under high-frequency data spikes, lazy-loaded modules, and Entra ID RBAC-gated views tracking LLM service latency and AIOps signal health org-wide.

Software Engineer - API Development & Cloud Security
Hanu Software (acquired by Insight Enterprises)

Jan. 2022 – Jan. 2023
Remote

- Sustained sub-100ms p95 latency under millions of high-concurrency enterprise API requests by applying core algorithms to optimize C++ and C# .NET hot paths, tuning CosmosDB indexing structures for distributed access, and enforcing OAuth 2.0 and RBAC authorization policies across Entra ID service modules
- Drove critical high-severity SAST security findings to zero and raised unit test coverage by 21% by engineering GitHub Actions CI/CD pipelines with Semgrep SAST gates, Gitleaks secrets scanning, and SonarQube enforcement templates adopted across 4 additional service teams.
- Cut Azure infrastructure spend by 18% over 3 months by executing Databricks Spark analytics on raw Cost Analysis telemetry to right-size VM instances and idle AKS node pools via Terraform automated provisioning with VMSS autoscaling, while improving overall CI/CD deployment pipeline velocity by 27%.
- Built a React 18 and TypeScript internal tooling dashboard powered by live REST API feeds, interactive Recharts visualizations, and secure Entra ID OAuth 2.0 PKCE authentication, delivering real-time RBAC-gated SAST vulnerability profiles and FinOps efficiency metrics to Security and Engineering leaders.

Software Engineer Intern - Data & Backend Infrastructure
Ernst & Young

May 2021 – Jul. 2021
Remote

- Streamed 1,842 daily industrial IoT sensor signals through MQTT broker ingestion into a 3-topic Apache Kafka pipeline on AWS, enabling high-throughput, low-latency real-time delivery to multiple distributed downstream consumer groups.
- Persisted time-series sensor telemetry to TimescaleDB on PostgreSQL, decoupling the durable analytical database store from the real-time Kafka path to consistently keep operational reads sub-second under heavy continuous ingestion
- Cut KPI dashboard query latency by 85% (from 250ms to 38ms) by engineering a Python and SQL Redis caching layer over Apache Spark aggregations, significantly reducing analytical database overhead on high-frequency metrics.

PROJECTS

go-pubsub-server *Go, Goroutines, RWMutex, REST API, WebSocket, PostgreSQL, Prometheus, Bearer Auth, Token Bucket Rate Limiting, Docker*

[Link](#)

- Engineered a high-throughput pub/sub server in Go achieving 1.54M msg/sec via goroutine-per-subscription fan-out, RWMutex topic registry, and token bucket rate limiting per API key at-least-once delivery guaranteed with DLQ after retry exhaustion, race detector clean under concurrent load.
- Designed dual-plane architecture (REST control, WebSocket data) with Bearer auth, PostgreSQL persistence via store interface, Prometheus metrics, and Go client SDK mirroring cloud.google.com/go/pubsub — enabling drop-in migration via config change.

Chronos v2 *Python, FastAPI, asyncpg, TimescaleDB, PostgreSQL, Redis, WebSocket, SSE, JWT RS256, Docker*

[Link](#)

- Built a time-series metrics API on TimescaleDB with hypertable partitioning, 7-day chunk compression, and continuous aggregates; engineered multi-tier Redis caching (60s/300s TTL) to reduce database load, served via FastAPI async endpoints with WebSocket and SSE feeds.
- Implemented JWT RS256 authentication with Redis-backed refresh token rotation, asyncpg connection pooling, and fine-grained RBAC roles validated by 80 passing integration tests covering auth flows, data access, and aggregation correctness across the stack.

TECHNICAL SKILLS

Languages & Protocols | Python, Java, Go, C++, C#, TypeScript, JavaScript, SQL, Bash, Protobuf, gRPC, REST, WebSockets, OAuth 2.0

Frameworks & Core | Java Spring Boot, FastAPI, .NET Core, Node.js, React, Angular, Resilience4j, Concurrency, Microservices

Data & Streaming | PostgreSQL, TimescaleDB, Redis, CosmosDB, DynamoDB, MongoDB, Apache Kafka, Event Hubs, Spark, asyncpg

Cloud & DevOps | GCP (Cloud Run, Build, Vertex AI), AWS (Lambda, EKS, SageMaker), Azure (AKS), Docker, Kubernetes, Terraform, Helm

Observability & Reliability | Prometheus, Grafana, OpenTelemetry, Distributed Tracing, SLO Ownership, Incident Triage, DRI Runbooks, CI/CD

AI Systems & Tooling | Multi-Agent Orchestration, MCP Server, LLM-as-a-Service, Ragas, Vector Search, Claude Code, Cursor, FinOps